Timothy Walker

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Rutgers University graduate with a B.S. in Computer Science. I am well versed in full-stack web development, application development, and object oriented programming. I have a strong passion for computing and a growing desire to expand the breadth of my knowledge.

EDUCATION

Rutgers University - New Brunswick NJ **2022**
Bachelor of Science (B.S.) in Computer Science | Major GPA 3.29 | Cumulative GPA 3.06

TECHNICAL SKILLS

Programming Languages: Java, C#, Javascript, Python, HTML, CSS, SQL
IDEs: Eclipse, Visual Studio, Visual Studio Code, Pycharm, Android Studio
Databases: MySQL, MS SQL Server
Other: GitHub, .NET, ASP.NET, REACT, Node.js, T-SQL, Azure Data Studio, REST APIs, TensorFlow

EXPERIENCE

AI Training Consultant | Outlier **January 2024 - June 2025**

- Optimized generative AI language models by solving complex coding tasks and refining AI responses.
- Developed efficient, functional code to enhance AI training and improve response accuracy.
- Evaluated and critiqued AI-generated outputs to ensure adherence to formatting, efficiency, and ethical standards.

PROJECTS **Links**

Full-Stack Development | Mechanical Keyboard Switch Database **Github | Deployment**

- Developed a full-stack web application to help keyboard enthusiasts navigate a custom database of mechanical keyboard switches with varying specifications that affect the typing feel and sound profile.
- Designed a responsive **React** frontend with dynamic filtering, gallery-style browsing, detail views, and state-driven UI updates.
- Includes real-time searching, seamless navigation, and dynamic routing with **REACT Router** for an intuitive browsing experience.
- Implemented a **RESTful ASP.NET Core API** backend powered by **Entity Framework Core** and **SQL Server**, enabling structured data management and flexible query handling for switch specifications.
- Web application, API, and database server all hosted through **Microsoft Azure**.

Embedded AI / Computer Vision | 3D Printer Failure Detection System **Github**

- Developed a real-time failure detection system for 3D printing on a Raspberry Pi with a camera module. Coded entirely in **Python**.
- Trained and optimized multiple **TensorFlow/Keras** models, based on **MobileNetV2**, to classify normal vs. failed prints, improving real-world detection with iterative tuning and validation.
- Implemented a lightweight, locally hosted, web interface to stream a live camera feed and process frames, outputting ‘Normal’ or ‘Failure’. Includes smoothing logic to **reduce false confidence**.
- Trained models can be converted to **TensorFlow Lite** for low-resource edge deployment.

Full-Stack Development | Mock Auction Website **GitHub**

- Developed a full-stack online auction platform supporting user bidding and item listings.
- Designed and implemented a **MySQL relational database** to manage users, auctions, and transactions.
- Frontend built with **HTML** and **Java Server Pages (JSP),** integrating **JDBC** for database connectivity.

Mobile and Desktop Application Development | Photo Album **GitHub**

- Developed two separate applications with identical functionality for managing and organizing photos.
- Built with **JavaFX** for desktop and **Android Studio (Java)** for mobile.
- Implemented image tagging, searching, and album organization features for an intuitive user experience.